eventually resulted in such features becoming stronger, weaker, bigger, or smaller in a species. For example, the ancestors of moles probably had good eyesight, but over generations this deteriorated because moles did not require vision as they burrowed underground. Similarly, giraffes gradually developed longer necks to enable them to reach leaves growing high up in trees.

“Nature is the system of laws established by the Creator for the existence of things and for the succession of creatures.”

The Comte de Buffon

Drivers of evolution

Lamarck’s ideas about inherited acquired traits were part of a wider early theory of evolution. He also believed that the earliest, simplest forms of life had emerged directly from nonliving matter. Lamarck identified two main “life forces” driving evolutionary change. One, he believed, made organisms develop from simple to more complex forms in a “ladder” of progress. The other, via the inheritance of acquired traits, helped them adapt better to their environment. When Charles Darwin developed his theory of evolution by means of natural selection, he would reject many of Lamarck’s ideas, but both men shared the belief that complex life evolved over an immense period of time.